



```
In [1]: from tensorflow.keras.models import Sequential
        from tensorflow.keras.layers import Embedding, LSTM, Dense, Dropout
        from tensorflow.keras.datasets import imdb
        from tensorflow.keras.preprocessing.sequence import pad_sequences

        import matplotlib.pyplot as plt
        import numpy as np
        import pandas as pd
```

```
In [2]: vocab_size = 10000

        (X_train, y_train), (X_test, y_test) = imdb.load_data(num_words=vocab_size)
```

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz>

17464789/17464789 ————— **6s** 0us/step

```
In [3]: max_length = 200

        X_train = pad_sequences(X_train, maxlen=max_length)
        X_test = pad_sequences(X_test, maxlen=max_length)
```

```
In [4]: LSTM_model = Sequential([
        Embedding(vocab_size, 128, input_length=max_length),
        LSTM(128, dropout=0.2, recurrent_dropout=0.2),
        Dense(1, activation='sigmoid')
        ])

        LSTM_model.compile(loss='binary_crossentropy', optimizer='adam', metrics=['acc'])
        history = LSTM_model.fit(X_train, y_train, epochs=5, batch_size=64, validation_data=(X_test, y_test))
```

Epoch 1/5

d:\anaconda\Lib\site-packages\keras\src\layers\core\embedding.py:100: UserWarning: Argument `input\_length` is deprecated. Just remove it.
warnings.warn(

391/391 ————— **96s** 240ms/step - accuracy: 0.7717 - loss: 0.4800 - val_accuracy: 0.7679 - val_loss: 0.5200

Epoch 2/5

391/391 ————— **94s** 239ms/step - accuracy: 0.8545 - loss: 0.3515 - val_accuracy: 0.8496 - val_loss: 0.3746

Epoch 3/5

391/391 ————— **97s** 249ms/step - accuracy: 0.8676 - loss: 0.3181 - val_accuracy: 0.8455 - val_loss: 0.3651

Epoch 4/5

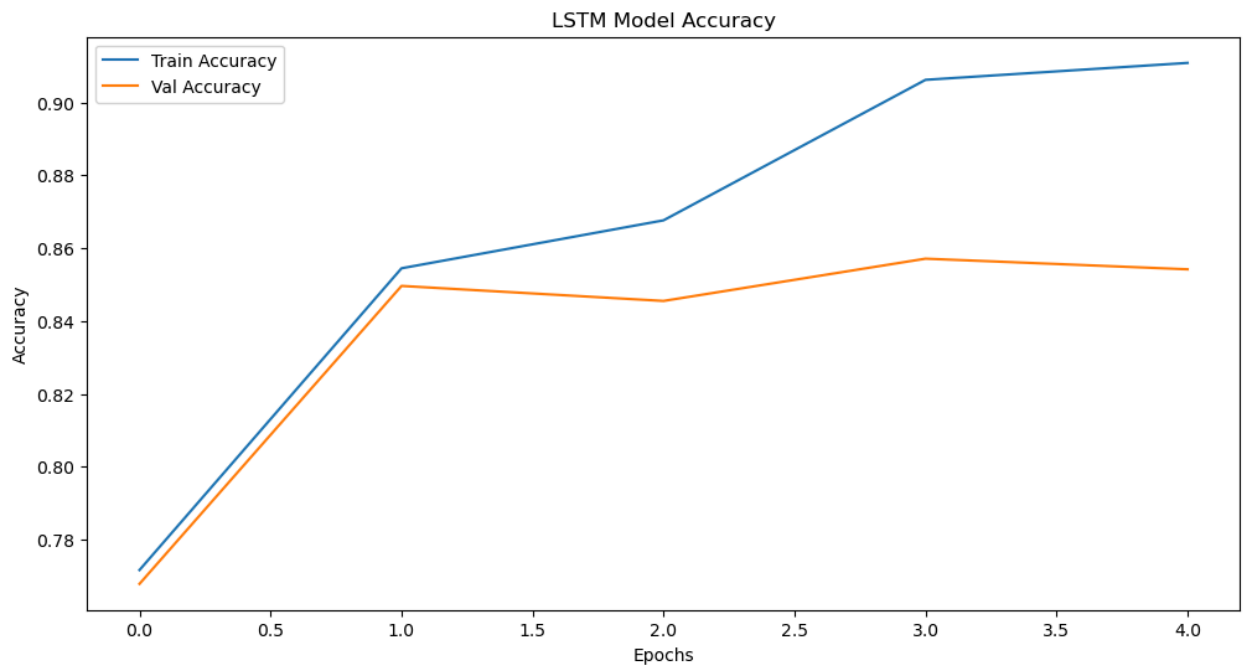
391/391 ————— **136s** 235ms/step - accuracy: 0.9062 - loss: 0.2412 - val_accuracy: 0.8571 - val_loss: 0.3972

Epoch 5/5

391/391 ————— **95s** 243ms/step - accuracy: 0.9108 - loss: 0.2247 - val_accuracy: 0.8542 - val_loss: 0.3798

```
In [5]: plt.figure(figsize=(12, 6))
        plt.plot(history.history['accuracy'], label='Train Accuracy')
        plt.plot(history.history['val_accuracy'], label='Val Accuracy')
```

```
plt.title('LSTM Model Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.show()
```



```
In [6]: plt.figure(figsize=(12, 6))
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.title('LSTM Model Loss')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.show()
```

